A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 31

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1** Very small gene pool so individuals are closely related and inbreeding is likely **1** Harmful recessive genes are expressed when both parents carry them: some goslings had thin hair-like feathers with poor insulation so survival in cold weather was low **1** A stud book records family trees so breeding is arranged between the least related individuals (often coordinated by a specialist zoo) **1** Exchange of individuals / semen between zoos – avoid hybridisation with related species / subspecies **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Breeding is triggered by stimuli (day length, light, temperature, food, body fat) which may be unknown / not provided **1 Population interactions:** mixing breeding and non-breeding individuals causes conflict – some need a choice of partners, others pair for life **1 Breeding habitat needed:** courtship display sites, nest sites, social groups (flamingos only breed in large groups - mirrors used) **1** Stress / disturbance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.3	**1 mark each** 1 mark for each of the 3 points below. **1 AI:** semen collected (and can be stored / cryopreserved for years) and inserted into the female, so animals need not be transported or meet – avoids dangerous mating (elephants, rhinos) and partner rejection **1 Embryo transfer:** a female of the endangered species (eg bongo) is treated with hormones to release many eggs, which are fertilised and implanted into surrogate females of a common related species (eland) **1** So many offspring are produced per cycle instead of one calf per year, overcoming the shortage of breeding females **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 33-35; AQA spec 3.1.2.3.3.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity **1** Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) **1** Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) **1** They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) **1** Many are in regions where environmental degradation / habitat loss threatens the wild plants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** A large population may all descend from a few closely related individuals so they are genetically similar **1** Similar adaptations mean all are susceptible to the same change (disease, climate) **1** Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs) – so species should be protected across their whole range to retain regional genes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
02.4	1 mark for the point below. **1** Morama bean (drought resistant, soya-like protein) – yeheb tree (drought resistant nuts, poor soils) – potato bean (high protein) – perennial maize / Kernza (no annual replanting, less ploughing / erosion) – salt-tolerant spinach relatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	
03.1	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Hedgerows **1** Hay meadows **1** Grazed moorland / heathland **1** Chalk grassland **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive **1** Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	2
03.3	1 mark for the point below. **1** The introduced (non-indigenous) predator may eat other species as well as the pest **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biological control** → using a natural predator, parasite or pathogen of the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	
04.1	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The size of the population, and the number of mature individuals able to breed **1** The rate of change of the population: how fast numbers have fallen, usually over the last ten years or three generations **1** The geographical range the species occupies and how far it is fragmented or dispersed, since a small or divided range makes extinction more likely **1** The quality and area of the remaining habitat, and whether the threats acting on the species are continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (12 000 - 2 100) / 12 000 x 100 **1** = 82.5% (accept 82 - 83%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks. • A bare wrong answer with no working.	2
04.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** It is a keystone species, so losing it would change the whole community and many other species would be lost with it **1** It is endemic to the area, so if it is lost there it is lost everywhere – a species found in many countries can be conserved elsewhere **1** It is a flagship or charismatic species that attracts public interest and funding, which also protects everything else living in its habitat **1** It is a genetic resource with known or likely value: a wild relative of a crop, or a source of a medicine **1** Conservation is practical and likely to succeed: the threats can be removed, and suitable habitat remains for a reintroduction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community • **endemic** → found naturally only in that one place **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 3 points below. **1** Appendix I lists species threatened with extinction, for which commercial international trade is banned and only exceptional permits such as scientific exchange are issued **1** Appendix II lists species that are not yet threatened with extinction but may become so unless trade is controlled, so trade is allowed under licence and monitored **1** Where the threat is demand in distant wealthy markets, cutting off the market removes the incentive to poach, and the agreement acts in every signatory country rather than only where the habitat is protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **CITES** → the treaty that controls international trade in endangered species **Do not credit** • 'CITES bans all trade in wildlife' - it controls and licenses trade, with different rules for each appendix	3
04.5	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Sites of Special Scientific Interest are notified for their wildlife or geology, and damaging operations on them need consent from the conservation agency **1** National Nature Reserves and Local Nature Reserves are owned or managed for conservation, with management plans and wardens **1** The Wildlife and Countryside Act makes it an offence to kill, injure or take listed species, or to damage their breeding sites and resting places **1** Special Areas of Conservation and Special Protection Areas, and Ramsar sites for wetlands, give international protection to habitats and to migratory birds **1** Green belts, National Parks and Areas of Outstanding Natural Beauty restrict development and so protect habitat indirectly **1** Agri-environmental schemes pay farmers to manage hedgerows, field margins and meadows for wildlife on land that is still farmed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-32, 38-39; AQA spec 3.1.2.3.1-3.1.2.3.2.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**1 mark each** 1 mark for each of the 2 points below. **1 Moral belief:** protecting the environment is the 'right' thing to do regardless of benefit to people **1 Enlightened self-interest:** protecting the environment is done because it causes fewer difficulties (for people) than not doing so **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Total impact = population size x the individual (per capita) impact of each person **1** Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.3	**Any 1 of 3** 1 mark each for any 1 of the 3 points below, in any order, maximum 1. **1** Slash and burn agriculture **1** Fishing **1** Discharge of sewage / biodegradable pollutants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **biodegradable** → broken down by decomposers **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Less dead organic matter / leaf litter so less humus and slower soil formation – fewer nutrients returned **1** Loss of canopy and litter exposes soil to rain splash and wind **1** Decomposing roots no longer bind soil so erosion, gullying and landslides increase (especially on slopes) **1** Reduced organic matter and biota lower infiltration and water retention **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	4
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind **1 Increased runoff:** more flooding after rain, lower flows in dry periods – sedimentation of rivers **1** Increased albedo of cleared land (reflects more sunlight) **1** Reduced carbon sequestration and loss of the carbon reservoir (CO_2 released by burning / decay) contributing to climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
06.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects **1** Loss of inter-species relationships (pollinators, seed dispersers) causing further declines **1** Loss of genetic resources / potential medicines before they are discovered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - radio waves do not travel through seawater – acoustic or satellite tags transmit position	1
07.2	**2 marks per point** 2 marks for each of the 2 points below; award 1 mark where a point is only partly made. **2** Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved **2** Radio / GPS tags or collars – leg rings / wing tags on birds – camera traps – DNA from droppings to confirm identity and diet **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
08.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 1000 / 100 000 x 100 = 1% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some light is reflected by the leaves **1** Some light is transmitted (passes through the leaves without being absorbed) or is of wavelengths not absorbed **1** Much of the energy captured is used in the plant's metabolic processes and released as low energy-density heat (respiration) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Less energy is passed to each trophic level than was received from the previous one (heat losses in respiration) **1** After four levels too little energy remains to support a further population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Permanent grassland is a more sustainable land use than arable crops in these areas (soil / rainfall unsuitable for crops) **1** Grazing animals can digest grass cellulose using symbiotic bacteria in their stomachs, producing food that humans can eat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
08.5	**1 mark each** 1 mark for each of the 2 points below. **1** Pigs are omnivores that can eat food wastes not wanted by people (eg cheese-making wastes, catering leftovers) **1** So edible meat is produced from material that would otherwise be wasted, without using land / crops that could feed people **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 4 points below. **1 Isolation makes colonisation difficult:** only marine species, seabirds, plants with floating seeds or chance arrivals (tortoises on logs, wind-blown birds) reach them **1** Whole taxa (terrestrial mammals, amphibians) may be absent so few original colonists **1** Colonists evolve in isolation under different selection pressures to fill the empty niches, producing unique species (Galapagos, Madagascar lemurs, Hawaii) **1** Endemics have small ranges and populations so are vulnerable to extinction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **endemic** → found naturally only in that one place • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** The birds evolved without mammal predators so nest on the ground / in burrows with no defence **1** Rats eat eggs and chicks – numbers fell to very low levels (45 and 12 pairs) **1** After eradication breeding success rose about 7-12 fold, confirming predation was the main limiting factor **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **limiting factor** → the factor in shortest supply that holds the rate or population back	3
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Introduced mosquitoes carry avian malaria to which endemic birds have no resistance – birds survive only above about 1500 m where it is too cold for malaria **1** Warming allows mosquitoes / malaria to spread to higher altitudes, shrinking the safe range **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **endemic** → found naturally only in that one place	2
09.4	1 mark for the point below. **1** Control of developments and visitor numbers (biosecurity, quarantine of imports) – protected areas – captive breeding of endemics **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **captive breeding** → breeding in captivity to increase numbers for release **Do not credit** • captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well	1

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 64-65; AQA spec 3.1.2.3.4.	
10.1	**1 mark each** 1 mark for each of the 2 points below. **1** t-test (or Mann-Whitney U if not normally distributed) **1** Comparing two means of counted / measured data from two independent samples **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **t-test** → a test for a difference between the means of two normally distributed sets • **Mann-Whitney** → a test for a difference between two sets of unpaired, non-normal data **Do not credit** • 'the result was significant' without comparing the calculated value with the critical value at p = 0.05	2
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** The difference is statistically significant (less than 1% probability that it arose by chance) **1** The null hypothesis (no difference between fields) is rejected – the organic field has significantly more earthworms **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **statistically significant** → a probability of 0.05, ie a 1 in 20 chance that the result happened by chance • **null hypothesis** → a statement that there is no significant difference or correlation **Do not credit** • 'the hypothesis was proved' - a statistical test either rejects the null hypothesis or fails to reject it	2
10.3	**1 mark each** 1 mark for each of the 2 points below. **1** A significant difference does not show a causal relationship **1** Other variables (soil type, moisture, pH, cultivation history) may differ between the two fields and cause the difference **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **causal** → one variable actually causing the change in the other **Do not credit** • 'organic farming uses no pesticides' - a restricted list of pesticides is permitted	2

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Scientific methodology:** identifying the topic, reviewing existing knowledge, formulating a testable hypothesis and null hypothesis, designing the study, analysing and drawing conclusions, planning further research • **Sample location:** random (grid coordinates / numbered sites) or systematic (transects, fixed intervals) to avoid bias – interval chosen so variation is detected without excess data • **Sample timing:** diurnal, seasonal, weather-related and long-term changes – sampling on different occasions – interval matched to rate of change • **Sample size and number:** larger and more samples for non-homogeneous variables – running mean and standard deviation show when the mean is reliable – preliminary study to decide • Standardisation and control of variables – accuracy, precision, reliability, representativeness – anomalies and repeats – calibration of equipment – appropriate techniques and risk management • **Statistical analysis:** choosing the test (Spearman, chi-squared, t-test, Mann-Whitney), significance and p-values, error bars, correlation vs causation • **Evaluation:** trade-offs between cost, time, disturbance and reliability – limitations of sub-sampling – justified conclusion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9

Do not write outside the box

Question	Marking guidance	Marks
	• **valid** → actually measures what it is meant to measure • **null hypothesis** → a statement that there is no significant difference or correlation • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • a correlation presented as proof of cause - say that it suggests a link that further work would have to test • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	
	Model answer - Level 3 (8 marks)	
	A valid investigation begins with a clear, testable hypothesis and a null hypothesis, informed by existing knowledge, so that the data collected can actually accept or reject it. The first design decision is sample location: random sampling (grid coordinates or numbered sites) or systematic sampling along transects avoids the bias that would come from choosing convenient sites, and a preliminary study shows the interval needed to detect variation without collecting unnecessary data.	
	Timing matters because many variables change diurnally, seasonally or with weather, so samples on different occasions are needed and the interval must match the rate of change. Sample size and number depend on how variable the population is: a running mean shows when enough samples have been taken and the standard deviation shows the scatter around the mean. Techniques must be standardised so results from different sites, times or researchers are comparable, other variables controlled or monitored, and equipment calibrated so that precise readings are also accurate; anomalies should be repeated rather than ignored.	
	Finally the analysis must be chosen to suit the data - Spearman's rank for correlation, chi-squared for frequencies, a t-test for two means or Mann-Whitney U for medians - so that significance ($p < 0.05$) can be assessed, remembering that a significant correlation does not prove causation. Every decision trades reliability against cost, time and disturbance; sub-samples can only estimate the whole, so the most valid study is the one whose sampling design makes its limitations explicit.	
	Weaker answer - Level 2 (5 marks)	
	When planning fieldwork you need a hypothesis. You should choose sites randomly so it is not biased, for example using random numbers. You need to take enough samples and repeat them so the results are reliable. The method should be the same each time. You should measure accurately with calibrated equipment. Then you do a statistical test such as a t-test to see if the results are significant. If p is less than 0.05 the result is significant. This makes the conclusions valid.	
	What makes the difference	
	• Level 2 is sound but generic: it says 'enough samples' where Level 3 explains how a running mean and standard deviation tell you how many.	
	• Level 3 covers timing and the choice between statistical tests; Level 2 mentions one test only.	
	• The evaluation is missing from Level 2 - there is no discussion of trade-offs or of correlation vs causation, so it cannot reach the top level however accurate it is.	

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.1, 3.7.2.3.	
11.1	**Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Species selection:** market, range of tolerance, controllability • **Abiotic control:** temperature, light, water, soil fertility / nutrients, aeration, salinity, pH, CO_2, topography, relief, wind - methods and examples • **Biotic control:** pest groups, cultural control, biological control (successes and failures), sterile males, pheromones, genetic resistance, pesticides (properties), antibiotics, pollinators, soil biota • **Manipulation of food species:** population density, monocultures, selective breeding, asexual reproduction, cross-breeding, AI / embryo transfer, GM – energetics - intensive vs extensive, energy subsidies, ratios, FCR, food chain losses • **Impacts:** habitat loss / change, reduced biodiversity, introduced species, pesticide and nutrient pollution, GHGs, hydrological changes, soil erosion – positive: plagioclimax habitats • Social, economic and political influences shaping methods • Sustainable strategies and evaluation	25
	– conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-337; AQA spec 3.5.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Physical features:** mass / gravity retaining an atmosphere and giving pressure for liquid water – distance from the Sun and the Goldilocks zone – speed of rotation reducing temperature extremes – tilt producing seasons – magnetic field deflecting the solar wind – the Moon stabilising the orbit and producing tides • **Importance of liquid water:** solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • **Development of life:** Archaea at vents – photosynthesis – oxygen reacting with iron then accumulating – ozone layer allowing colonisation of land – aerobic organisms • **Life changing conditions:** carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases	25
	• **Methods of studying past conditions:** proxy data, isotope ratios, ice cores, Mg:Ca ratios – understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	

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